MODULATOR BIAS CONTROLLER
MBC-DG-LAB

USER MANUAL

Version 1.4

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Legal Notice 3
Disclaimers 3
About this Manual 3
Purpose 3
Audience 3
Prerequisites 3
Terms and Symbols in this Manual 3
Electrical Safety 4
Ventilation of the MBC-DG-LAB 4
Contacting exail PSD: 4
General Information 5
Mach-Zehnder intensity modulators 5
Product Overview 6
Principle 6
Mechanical Dimensions 8
Getting started 9
Standard Accessories 9
Installation and first time operation 9
Operation 11
Selecting Auto/Manual mode 11
Selecting transfer level 11
Auto Calibration 11
Adjusting dithering frequency 11
Adjusting dithering amplitude 11
Adjusting the photocurrent gain circuit 11
Troubleshoot: 12

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About this Manual

Purpose

The purpose of this instruction manual is to explain how to set up the MBC-DG-LAB and to utilize the various features of the product.

Audience

This manual is designed to professionals with the necessary technical background and prerequisites needed to use fiber optics components, including Mach-Zehnder lithium niobate modulators, and the related test equipments and accessories.

Prerequisites

Reader is assumed to be familiar with fiber optics technology.

Terms and Symbols in this Manual

The following terms and symbols may appear throughout this manual :

WARNING : Warning statement identify conditions or practices that could result in injury or loss of life

CAUTION : Caution statement identify conditions or practices that could result in damage to this product or other property

Electrical Safety

	WARNING	**Do not attempt to perform servicing or maintenance** To avoid personal injury, do not operate this instrument without the protective cover of the chassis. Do not make any service or maintenance of any kind to the system. Refer servicing only to authorized personnel from exail.
	WARNING	**Operate under the proper environmental conditions.** The MBC-DG-LAB is designed for indoor use, only. To avoid the possibility of injury, do not expose this instrument to rain or excessive moisture. Do not operate the instrument in the presence of flammable gases or fumes.

Ventilation of the MBC-DG-LAB

	WARNING	Do not cover the MBC-DG-LAB with any covering or overlay.

Contacting exail PSD:

For application oriented questions, for questions related to service, please contact exail PSD Besançon

General Information

Mach-Zehnder intensity modulators

Lithium niobate (LiNbO3) Mach-Zehnder intensity modulators are external modulators that have been widely used since the mid-nineties, mostly in long haul optical networks for their ability to modulate optical signal from 1300 nm and 1550 nm laser diodes with short transition times and without chirp. Thanks to their broad wavelength performance, they currently also offer modulation solutions to 1064 nm and other near infrared lasers.

Mach-Zehnder modulators are interferometric devices and exhibit a sine transfer function (Figure 1). Although highly stable components, their operating point can suffer slow drift due to variations of external conditions and that drift may result in variations of key performances like extinction ratio.

In order to adjust the operating point of Mach-Zehnder modulators independently from the high frequency modulation signal applied, they can be designed with two sets of electrodes: one set of electrodes, the RF Electrodes, is used to apply the RF signal (PRBS data stream, analog signal, short pulses…). The second set of electrodes, the DC Bias Electrodes, is generally used to adjust with a fixed voltage the working point of the modulator. Figure 2 illustrate the typical electrode architecture of Mach-Zehnder modulator.

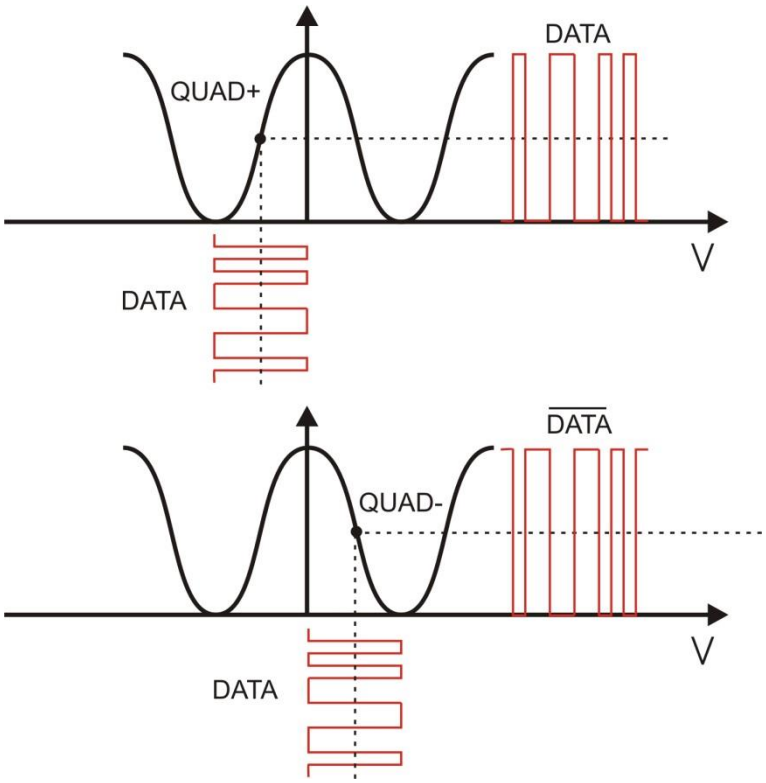


Figure 1 : transfer function of a Mach-Zehnder modulator and data output vs data input at QUAD+ and QUAD- operating point

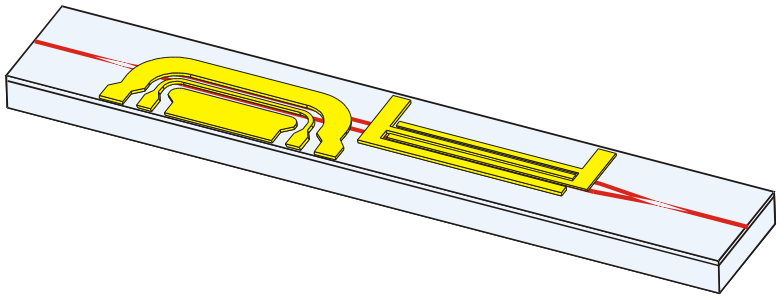


Figure 2 : typical electrode and waveguide architecture of a Mach-Zehnder modulator

Product Overview

MBC is a bias controller specially designed to stabilize the operating point of LiNb03 Mach-Zehnder modulators by monitoring the bias voltage applied on the DC electrodes of the devices.

The MBC controllers can lock any modulator and allows for continuous tuning of the operating point. The continuous format such as ERZ (Electrical Return to Zero) where the operating point must be set at 33% or 66% of the maximum transmission.

Principle

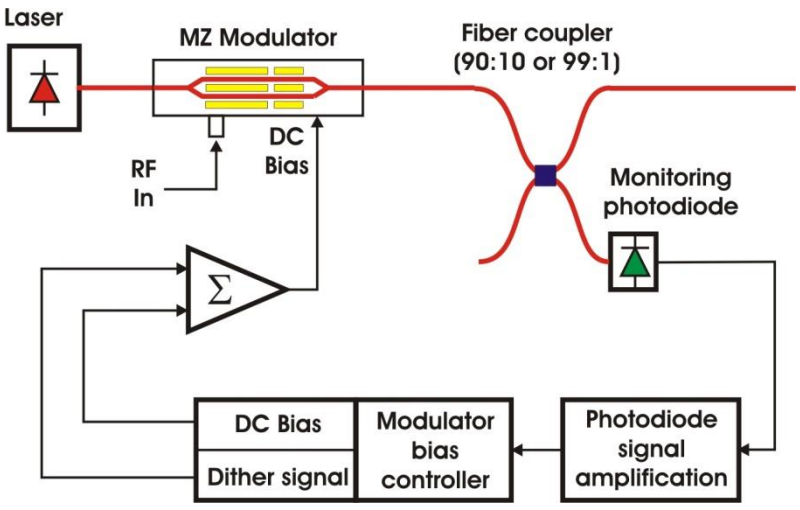
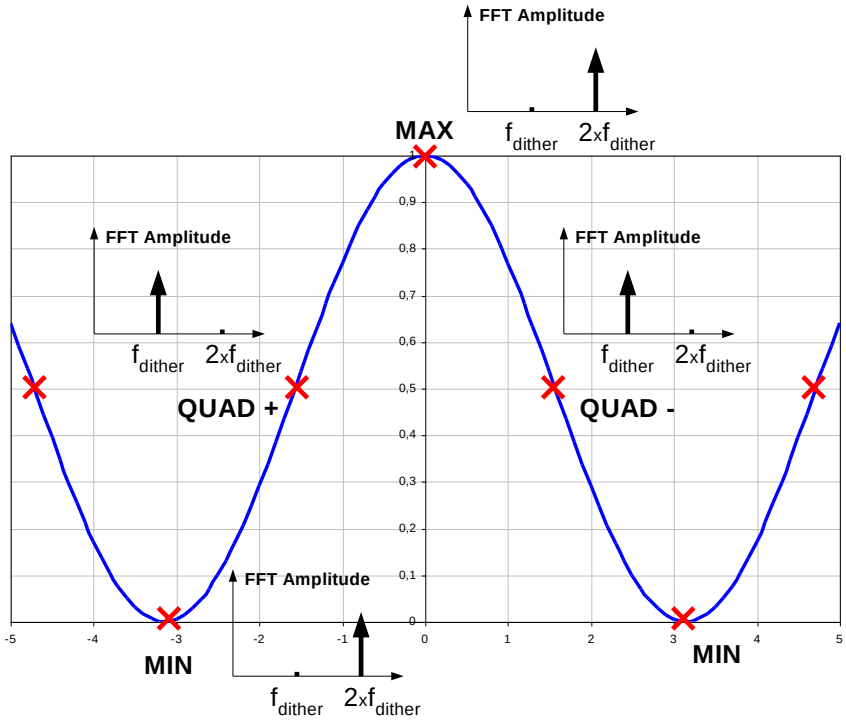


Figure 3: basic scheme of a modulation set-up including a modulator bias controller and an external monitoring photodiode

A Mach-Zehnder modulator is illuminated by a laser at the optical input. The data stream is applied to the RF electrodes. At the output of the MZ modulator, a fiber coupler with a strong imbalance (10:90 or 1:99) allows to detect a small portion of the transmitted light with a monitoring photodiode. The signal is amplified and processed in order to control the stability of the operating point. Any deviation is compensated by a new value of the electrical bias voltage applied by the feedback loop in order to maintain the operating point at the desired position.

The principle of operation of the MBC is based on the generation of a very low frequency and low amplitude electrical signal (dither signal), which is summed to the DC bias voltage. Any deviation from the working point creates harmonics that can be measured and compared by frequency analysis and by signal processing carried out by a specific algorithm, developed by exail, in order to retrieve the sign and the amplitude of the deviation.

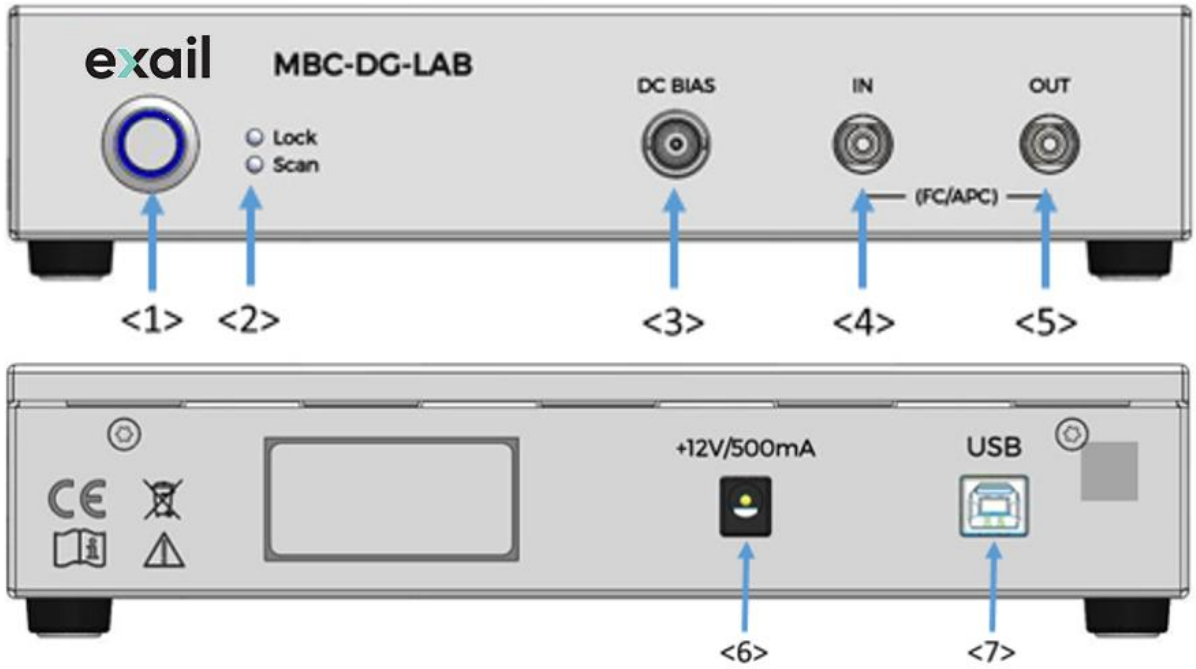
The modulator is polarized in quadrature QUAD+ or QUAD- as the second harmonic is null. Otherwise, the modulator can be polarized on MIN or MAX as the first order is null. Any other transmission state is also possible thanks to a dedicated algorithm function.



Mechanical Dimensions

221mm x 221mm x 44mm

MBC LAB:



<1> Power button

<2> Control LED
Lock: MBC is locked to the fixed point (still active when laser light is off)
Scan: MBC is performing a scan. If flashing, no signal detected.

<3> Bias Modulator connector (BNC)

<4> Input Optical Connector

<5> Output optical connector (coupler option)

<6> Power supply connector (Jack 3.5mm, 12V 500mA)

<7> USB connector

Getting started

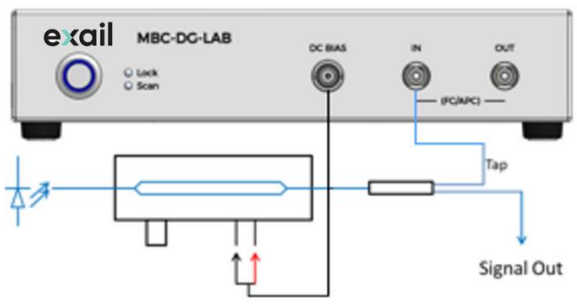
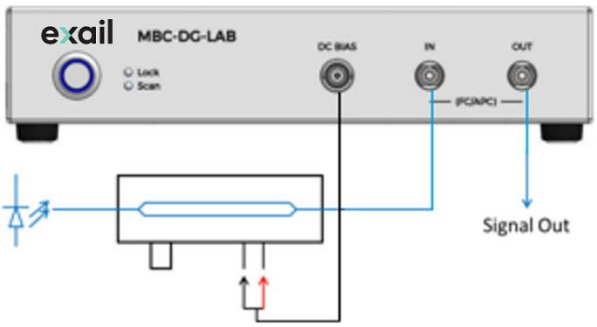
Standard Accessories

User Manual
Interface User Manual
Cord Kit
USB Cord
AC-DC Power Supply

Installation and first time operation

Proceed as follow to install the MBC-DG-LAB:

- Set MBC LAB on a flat stable surface
- Connect the power supply plug-in to the power supply connector <6>
- Connect the modulator optical output through a tap coupler for "no coupler option", directly for "coupler option" to "In"-connector of the MBC-Lab(<4>).See optical connection schemes bellow.
- Connect the BNC cable to "DC-Bias" (<3>) and the black (Ground) and red (DC Bias) wires to the bias electrode of the modulator.
- Switch on the laser to illuminate the modulator at the desired optical power level
- Power ON the MBC-DG-LAB <1>
- A procedure of initialization by scanning the Vbias range from –10V to +10V will start to adjust automatically the operating point by use of the initial conditions
- After stabilization, the RF signal can be applied and the parameters of the MBC can be modified.
- Any change of the dither frequency results in a new search of the operating point by scanning of the bias voltage from –10V to +10V.

Setup with an external coupler (no coupler option):	**Setup with an internal coupler (coupler option):**
	

IMPORTANT NOTES:

1- The optical signal processing in the MBC is such that the value of the RF amplitude voltage is limited. The limit is 0.75Vπ and the MBC will not work correctly if the RF signal with a larger amplitude is applied to the modulator.
A 0.75Vπ amplitude is equivalent to a 1.5Vπ peak-to-peak value and the majority of the applications operate with a lower RF signal: typically 0.5Vπ amplitude (or 1Vπ peak-to-peak). Thus, in spite of that restriction, the MBC will operate correctly with most of the application.

Operation

For more information on the way to set parameters, please refer to the GUI interface handbook.

Selecting Auto/Manual mode

One can choose between two operation modes: AUTO or MANUAL.
In AUTO mode, the board searches the bias value corresponding to the selected transfer level.
In MANUAL mode, one set the DC bias value, and no dither is applied to DC Bias.

Selecting transfer level

Transfer Level is characterized by its level in % (MIN is 0, MAX is 100, QUAD+ is +50, QUAD – is –50). In general for digital NRZ and analog signals, the transfer level is set to QUAD. For pulse application, MIN point is privileged.

Auto Calibration

Auto Calibration will set the parameter dither amplitude and photodiode gains to an optimum, depending on the chosen transfer level point: Quad, Min or Max. The board will switch to Auto mode at the end of the calibration sequence.

Adjusting dithering frequency

The MBC allows the user to adjust the dithering frequency between 400 Hz and 1400 Hz. It is set by default at 1000Hz. This feature proves especially useful to avoid any interference between the RF signal and the dithering signal or when more than one modulator with MBC are used in the same link.

Adjusting dithering amplitude

The MBC allows the user to adjust the dithering amplitude between 10 mV and 1 V. This feature proves especially useful to optimize the overall performance of the MBC and to adapt to various conditions such as modulator Vpi or photodiode response. In auto calibration mode dither amplitude is set to 3% of the Vpi in Quad mode and 0.2% in Min and Max mode. Increasing dither, in general 20 to 50mVpp can help for feedback loop stabilization at low optical power (-20dBm on photodiode for example).
After modifying this parameter, one must do a scan restart.

Adjusting the photocurrent gain circuit

The MBC 2 offers 127 different gain values for the photocurrent circuit so as to adapt to the largest variety of applications and to offer the widest dynamic range for the input photocurrent. For Quad mode it is recommended to keep the gain under 110, otherwise the feedback loop may oscillate.
After modifying this parameter, one must do a scan restart.

It must be noticed that the photocurrent range within a gain position may slightly vary with the operating mode and the dithering amplitude. The Min mode requires less photocurrent than the Quad+ mode due to the higher sensitivity of the detection. Also, higher dithering amplitude will make the detection easier and will lower the low limit of the photo current range.

Troubleshoot:

Shifted Quad point	Quad point can be shifted, when the optical power illuminating the photodiode is too high. It will happen when the photodiode current increase over 500µA at max point. Optical power onto the photodiode has to be decreased.
Feedback loop oscillation	If the feedback loop oscillates fast, one should increase the dither amplitude (10 to 50mVpp) and reduce the photodiode gain (10 to 20). If it doesn't work, one should check that the optical power on the photodiode at MAX point is sufficient to obtain a minimum of 8.5µA at the output of the photodiode.